

Rolling Stock Scheduling

Mathematical Formulation

Problem Overview

Rolling stock scheduling assigns a fleet of vehicles (units) to a fixed timetable of trips so that every trip is adequately served, while minimizing operating cost and respecting physical, temporal, and maintenance constraints. Structurally, this is a **Vehicle Routing Problem with Time Windows (VRPTW)**, extended with a resource-reset constraint (maintenance) and a multi-vehicle-per-job constraint (coupling). It is NP-hard.

1 Sets

- U — set of rolling stock units
- T — set of trips
- S — set of stations

2 Parameters

Trip parameters (for trip $i \in T$)

- $o_i, d_i \in S$ — origin and destination station
- dep_i, arr_i — scheduled departure and arrival time
- τ_i — minimum turnaround time required at d_i before reuse
- ℓ_i — trip distance (km)
- ρ_i — expected passenger demand
- κ_i — maximum number of coupled units permitted on trip i

Unit parameters (for unit $u \in U$)

- c_u — passenger capacity
- $type(u)$ — rolling stock type (e.g. EMU4, EMU8)
- $h_u \in S$ — home depot
- a_u — earliest time available at h_u

- μ_u — accumulated kilometres since last maintenance, at start of day
- $M_{type(u)}$ — maximum permitted km between maintenance visits for that type

Network parameters

- $t(s_1, s_2)$ — travel time (minutes) between stations $s_1, s_2 \in S$
- $\delta(s_1, s_2)$ — travel distance (km) between stations $s_1, s_2 \in S$
- $F(s) \in \{0, 1\}$ — indicator that station s has a maintenance facility

3 Decision Variables

$x_{u,i} \in \{0, 1\}$	unit u serves trip i
$y_{u,i,j} \in \{0, 1\}$	unit u travels directly from trip i to trip j
$z_u \in \{0, 1\}$	unit u is activated (used at all) during the day
$m_{u,i} \in \{0, 1\}$	unit u undergoes maintenance immediately after trip i

4 Objective Function

$$\min \quad w_1 \sum_{u \in U} \sum_{i \in T} \delta_{\text{deadhead}}(u, i) x_{u,i} + w_2 \sum_{u \in U} z_u + w_3 \sum_{u \in U} \sum_{i \in T} E_{u,i} + w_4 \sum_{i \in T} \max \left(0, \rho_i - \sum_{u \in U} c_u x_{u,i} \right) \quad (1)$$

where $E_{u,i}$ denotes the excess kilometres unit u has accumulated beyond $M_{type(u)}$ at the point of trip i , and $\delta_{\text{deadhead}}(u, i)$ is the empty repositioning distance unit u must travel to reach trip i 's origin. The four weights $w_1, \dots, w_4$ trade off deadhead travel, fleet size, maintenance overruns, and unmet demand respectively.

5 Constraints

(C1) Demand coverage

$$\sum_{u \in U} c_u x_{u,i} \geq \rho_i \quad \forall i \in T \quad (2)$$

(In the relaxed/soft-constraint version used by the heuristic solver, violations are penalized rather than forbidden — see the shortfall term in the objective.)

(C2) Coupling limit

$$\sum_{u \in U} x_{u,i} \leq \kappa_i \quad \forall i \in T \quad (3)$$

$$\sum_{u: type(u)=\text{EMU8}} x_{u,i} \leq 1 \quad \text{and} \quad x_{u,i} = 0 \quad \forall u' : type(u') = \text{EMU8}, u \neq u', x_{u',i} = 1 \quad (4)$$

i.e. a fixed-formation type (EMU8) cannot be coupled with any other unit on the same trip.

(C3) Chain flow conservation

Each unit's assigned trips must form a single continuous path (no branching):

$$\sum_{j \in T} y_{u,i,j} = x_{u,i}, \quad \sum_{i \in T} y_{u,i,j} = x_{u,j} \quad \forall u \in U, \forall i, j \in T \quad (5)$$

augmented with a virtual depot-source arc into each unit's first trip and a depot-sink arc out of its last.

(C4) Temporal and spatial feasibility of a chain link

An arc $y_{u,i,j} = 1$ is permitted only if the unit can physically reposition and rest in time:

$$y_{u,i,j} = 1 \Rightarrow arr_i + \tau_i + t(d_i, o_j) \leq dep_j \quad (6)$$

This is the constraint that forbids a unit from being in two places at once.

(C5) Depot start condition

For unit u 's first trip i in its chain:

$$a_u + t(h_u, o_i) \leq dep_i \quad (7)$$

(C6) Maintenance threshold

Let $K_{u,i}$ denote unit u 's cumulative kilometres run up to and including trip i :

$$K_{u,i} = (1 - m_{u,i-1}) K_{u,i-1} + \delta(\text{prev. location}, o_i) + \ell_i \quad (8)$$

$$K_{u,i} \leq M_{type(u)} \quad \forall u \in U, \forall i \in T \quad (9)$$

$$m_{u,i} = 1 \Rightarrow F(d_i) = 1 \quad (10)$$

i.e. a maintenance reset may only occur when the unit's location has a maintenance facility.

(C7) Activation linking

$$z_u \geq x_{u,i} \quad \forall u \in U, \forall i \in T \quad (11)$$

Summary

Constraint	Meaning
C1	Every trip's passenger demand must be met by assigned unit capacity
C2	Coupling on a trip is bounded, and fixed-formation types cannot be coupled
C3	Each unit's trips form one continuous path, not branches
C4	A unit cannot start a trip before it can physically arrive there
C5	A unit's first trip must be reachable from its home depot in time
C6	A unit cannot run past its maintenance mileage threshold without a valid reset
C7	A unit counted as "activated" if it serves at least one trip